

X RAYS INTRODUCTION

3-D imaging tomography

X rays CT

PET

SPECT

MRI

Ultrasound

Optical and Fluorescence

OCT (Optical coherence tomography)

PET and SPECT

Injection of radioactive tracers

Expensive tracers with short half life

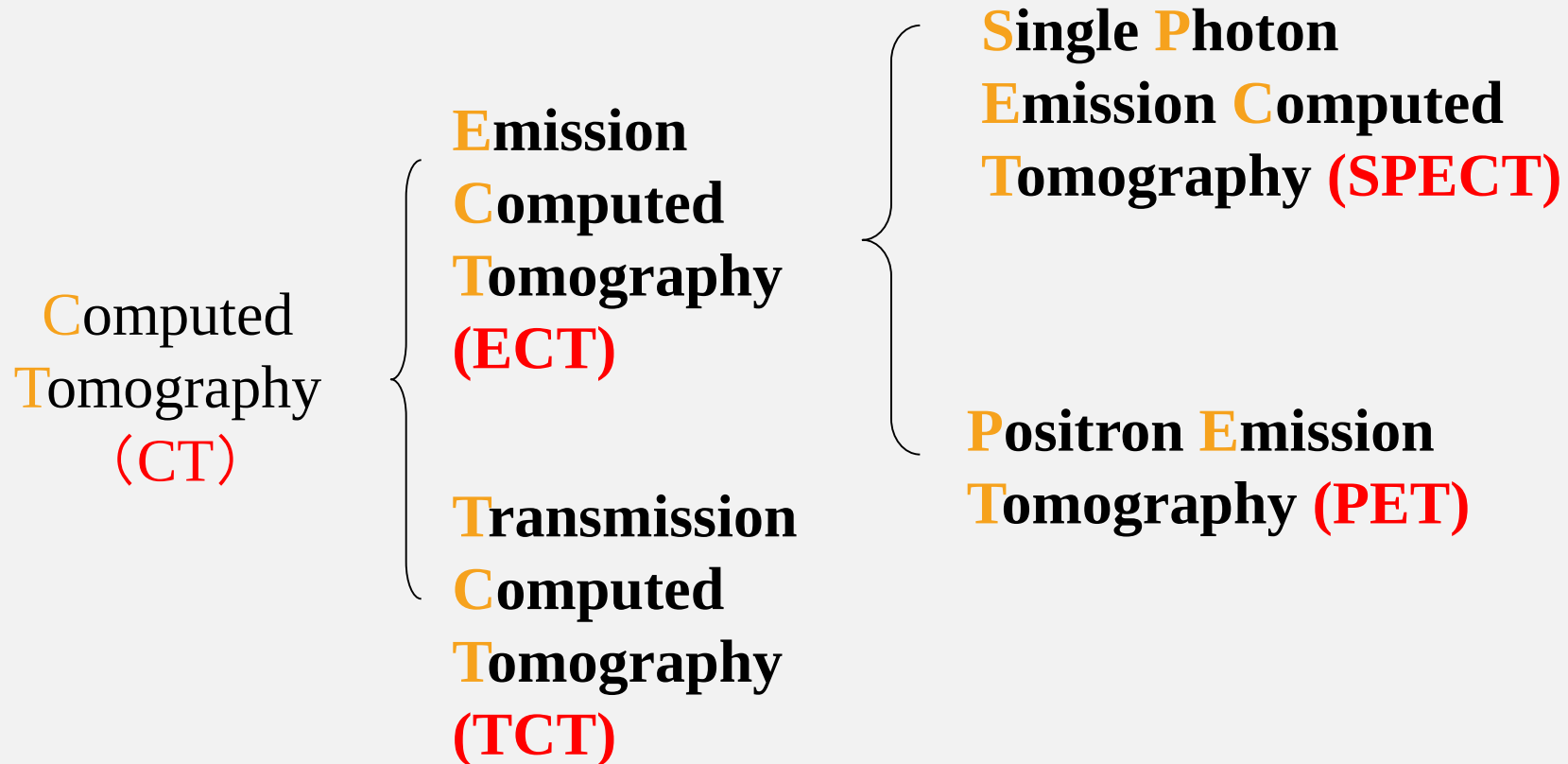
Diagnostic Imaging

➤ Transmission

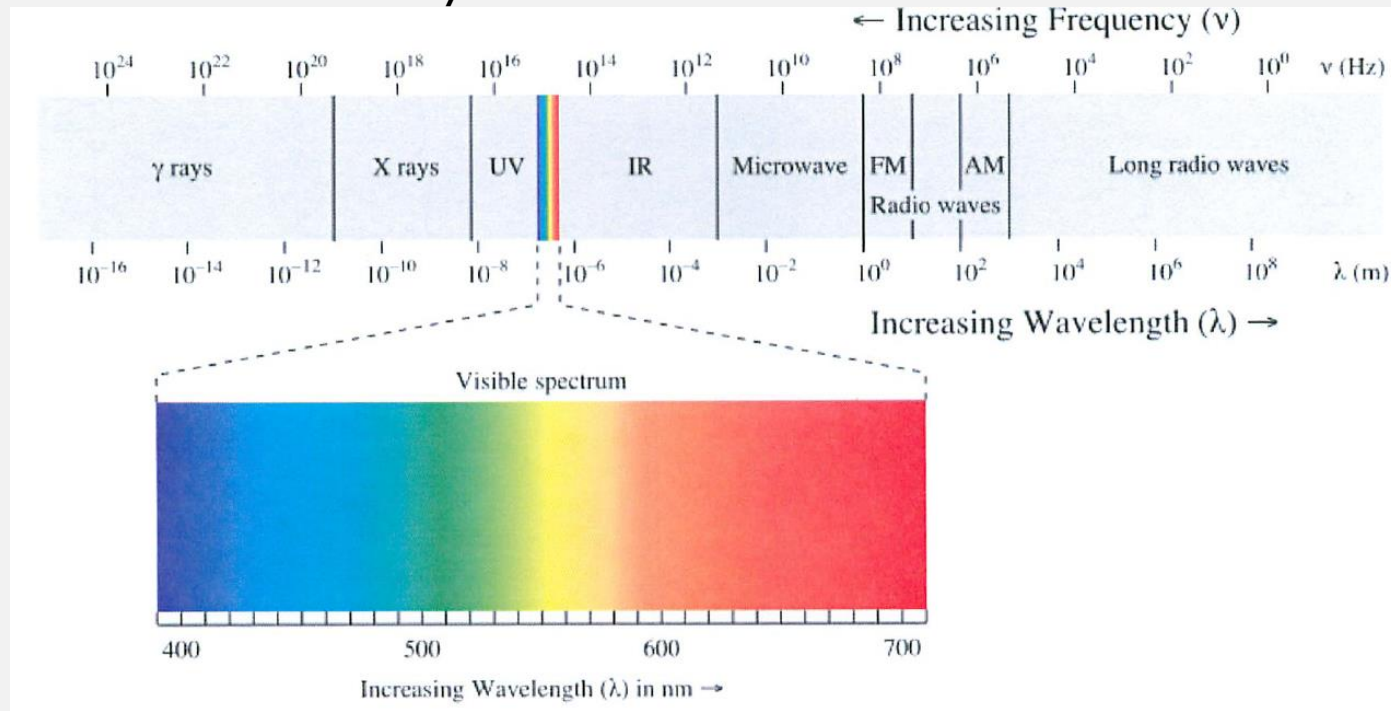
➤ Reflection

➤ Emission

Computed Tomography (CT)



X rays



CLASS	FREQUENCY	WAVELENGTH	ENERGY
γ	300 EHz	1 pm	1.24 MeV
HX	30 EHz	10 pm	124 keV
SX	3 EHz	100 pm	12.4 keV
	300 PHz	1 nm	1.24 keV
EUV	30 PHz	10 nm	124 eV
NUV	3 PHz	100 nm	12.4 eV
NIR	300 THz	1 μ m	1.24 eV
MIR	30 THz	10 μ m	124 meV
	3 THz	100 μ m	12.4 meV

Legend:

γ = [Gamma rays](#)

HX = Hard [X-rays](#)

SX = Soft X-Rays

EUV = Extreme [ultraviolet](#)

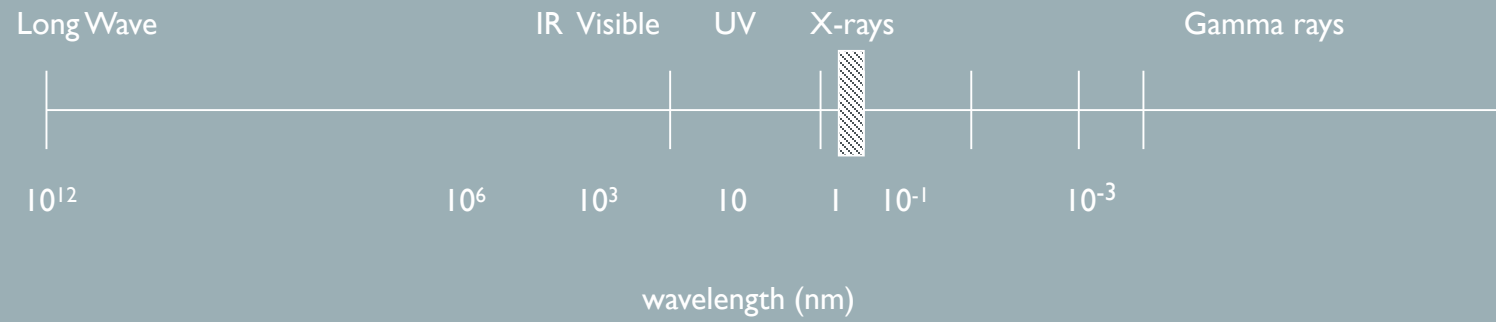
NUV = Near ultraviolet

[Visible light](#)

NIR = Near [infrared](#)

MIR = Moderate infrared

FIR = Far infrared



X-rays



Frau Röntgen's hand



X rays are electromagnetic rays

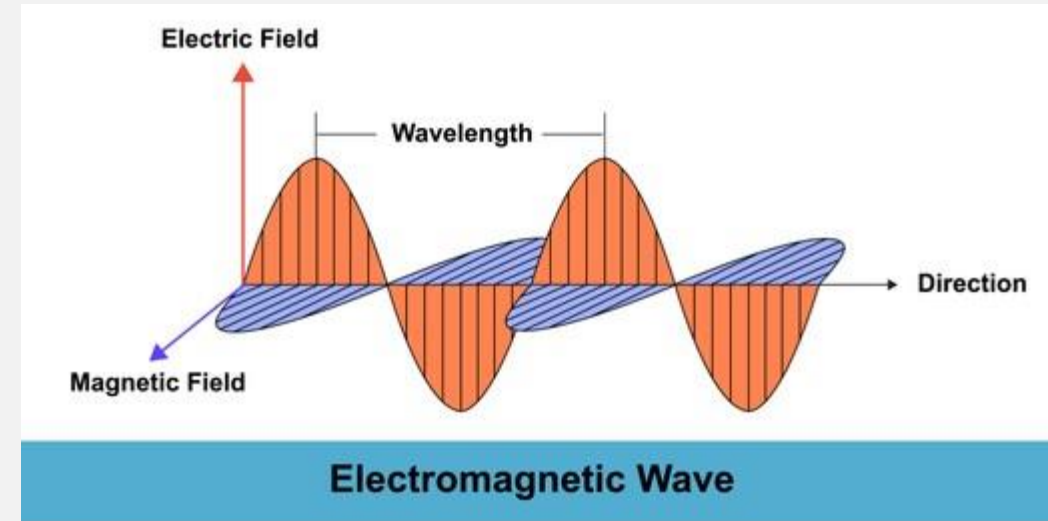
c : velocity of light 3×10^8 m/s

$$\lambda = c / \nu$$

$\lambda_0 = c / \nu$ Where c is velocity of light in vacuum

$$\lambda = \nu \cdot T \quad (\text{in Medium})$$
$$= \nu \cdot (1 / \nu)$$

$\lambda < \lambda_0$ wavelength in medium is less than wavelength in vacuum



C: velocity of light 3×10^8 m/s

Energy = $h\nu$ (h= planks const, 6.6×10^{-34} J s ν =frequency)

Visible light : imaging of scattered light

X rays: absorption of X rays in tissues

Technique based on Absorption

X rays: higher energy: Penetrate deep into tissue

Higher Energy : May damage tissue: Mutations etc

Broad Applications

Chest

Ribs, Lungs

Bones,

Fluoroscopy, Angiography, Mammography

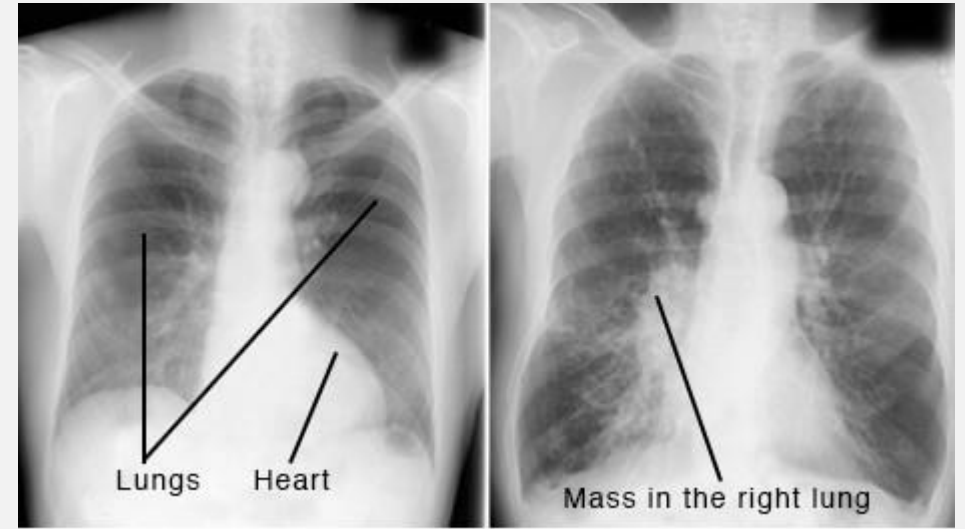
CT

Applications

Human Body: Chest, lungs, ribs, bones fractures etc

Dental applications

Fluoroscopy, Mammography, Angiography, CT

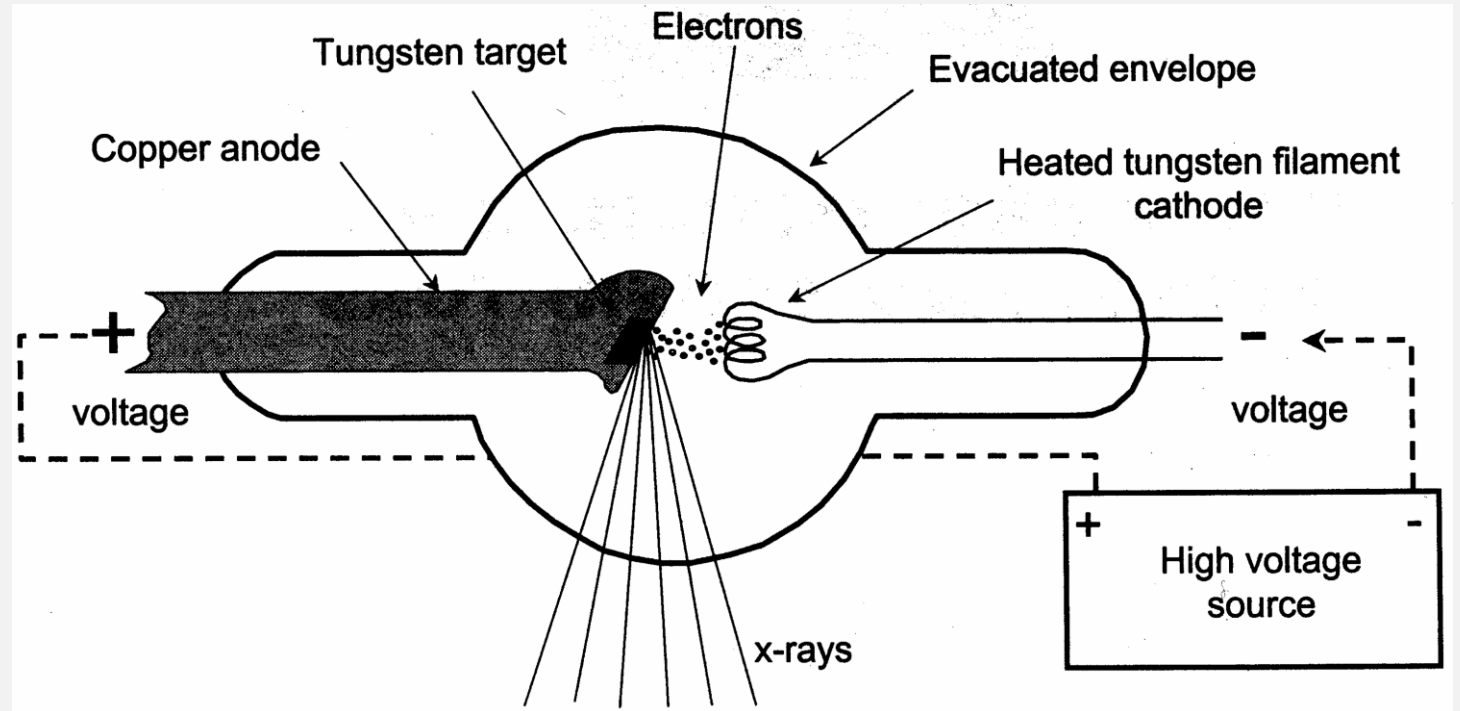


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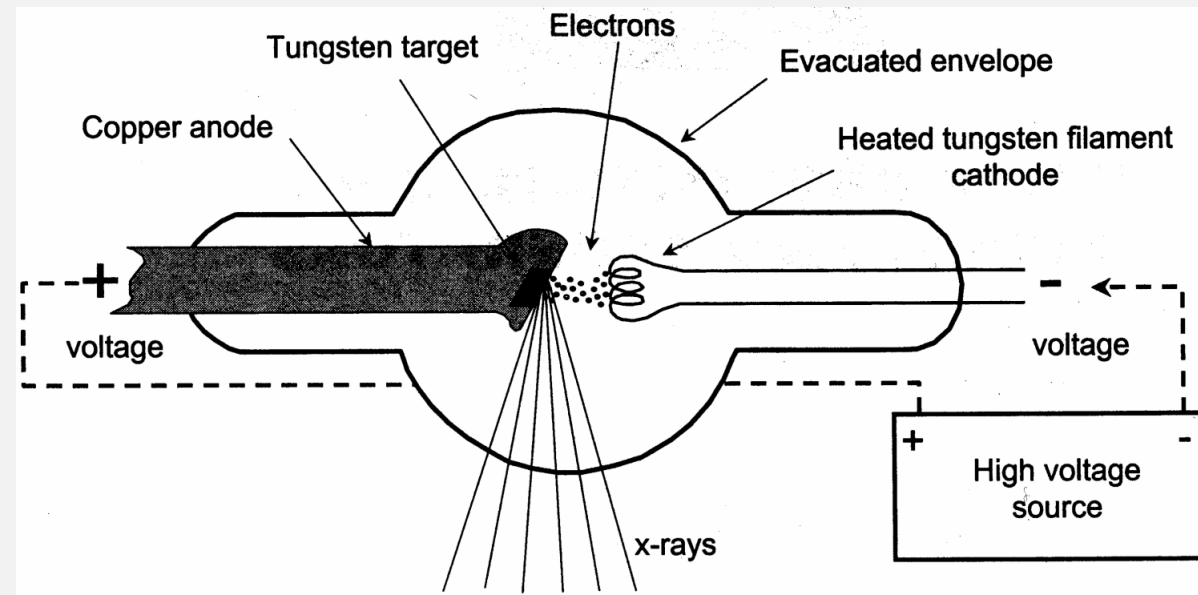
X ray generation

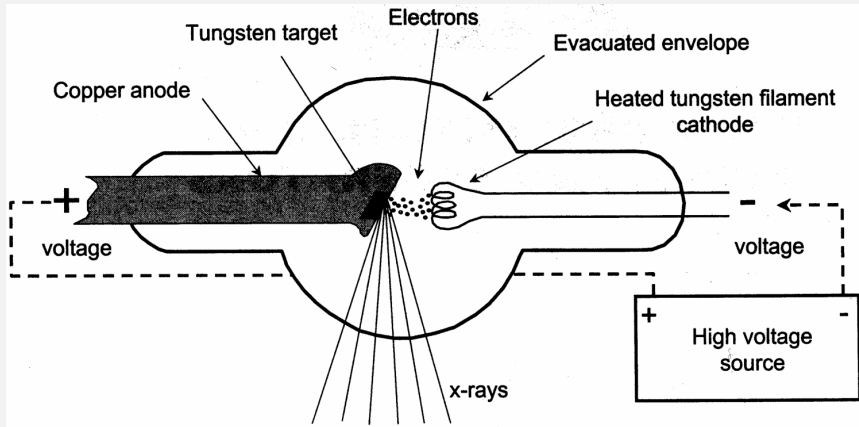
KE of Electrons transformed into electromagnetic waves to produce X rays



Bremsstrahlung

- A large potential difference is applied across the two electrodes in an evacuated (usually glass) envelope.
 - Negatively charged cathode is the source of electrons (e^-).
 - Positively charged anode is the target of electrons.
- Electrons released from the cathode are accelerated towards the anode by the electrical potential difference and attain kinetic energy.



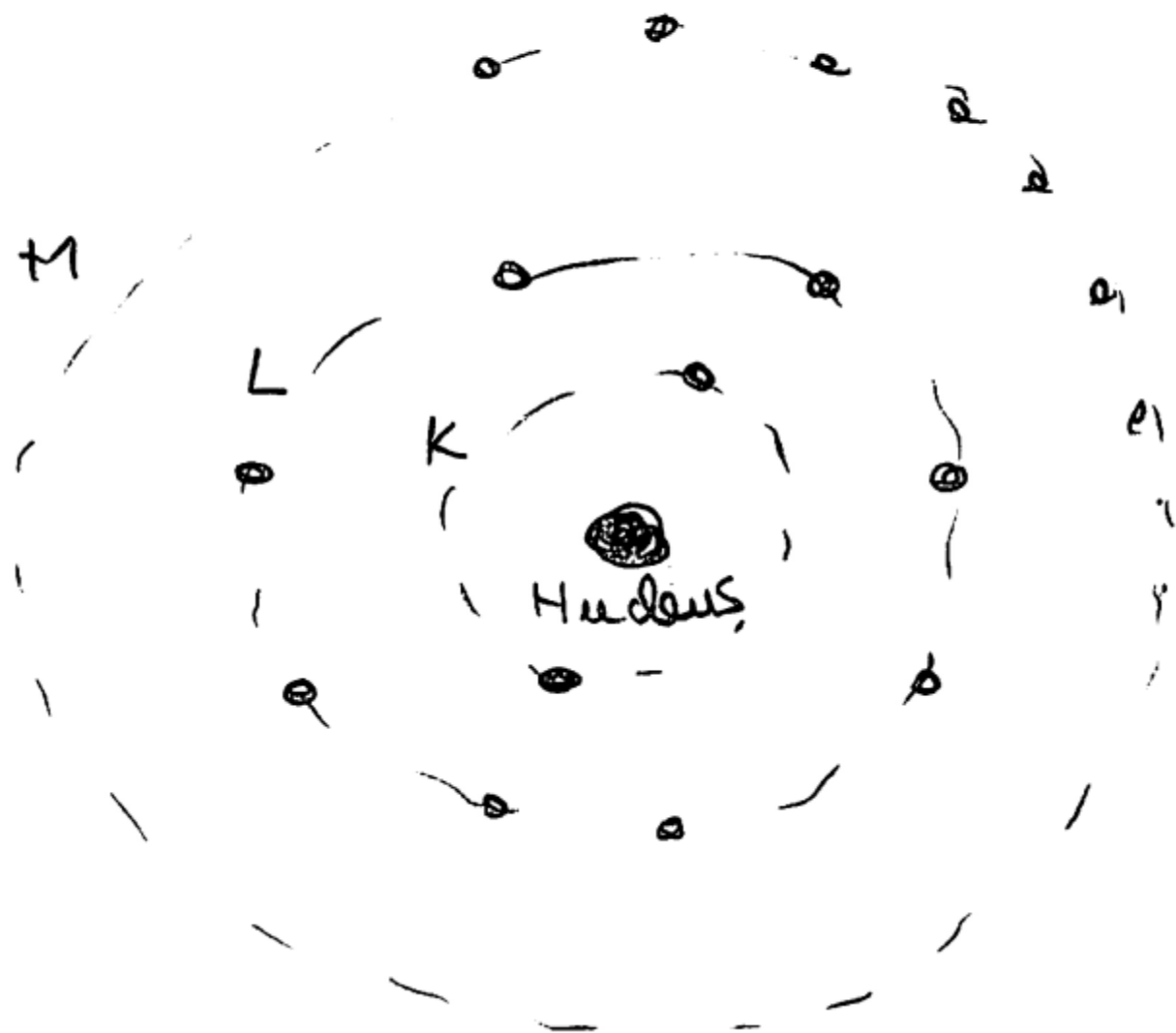


PD~150 KV Between Anode and Cathode
Electrons are accelerated to have High KE
Electron emission occurs from the Tungsten element held at 2200 deg C (thermionic emissions)
Rotation of Anode avoids overheating
Electrons are focused on to Anode
Ems penetrate about 10 microns

Anode

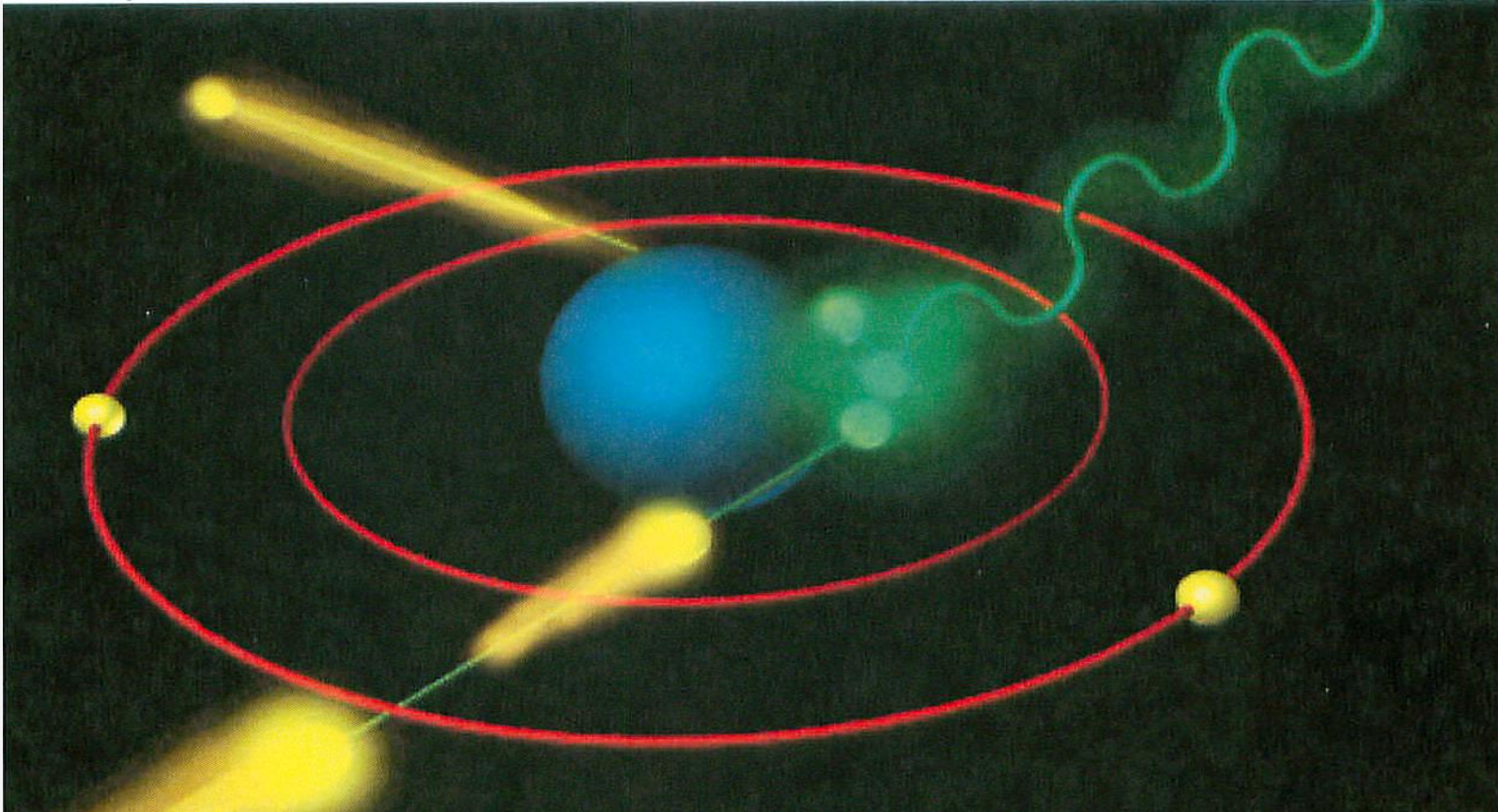
High melting point
High thermal conductivity
Low Vapor pressures
High Atomic Number → High efficiency

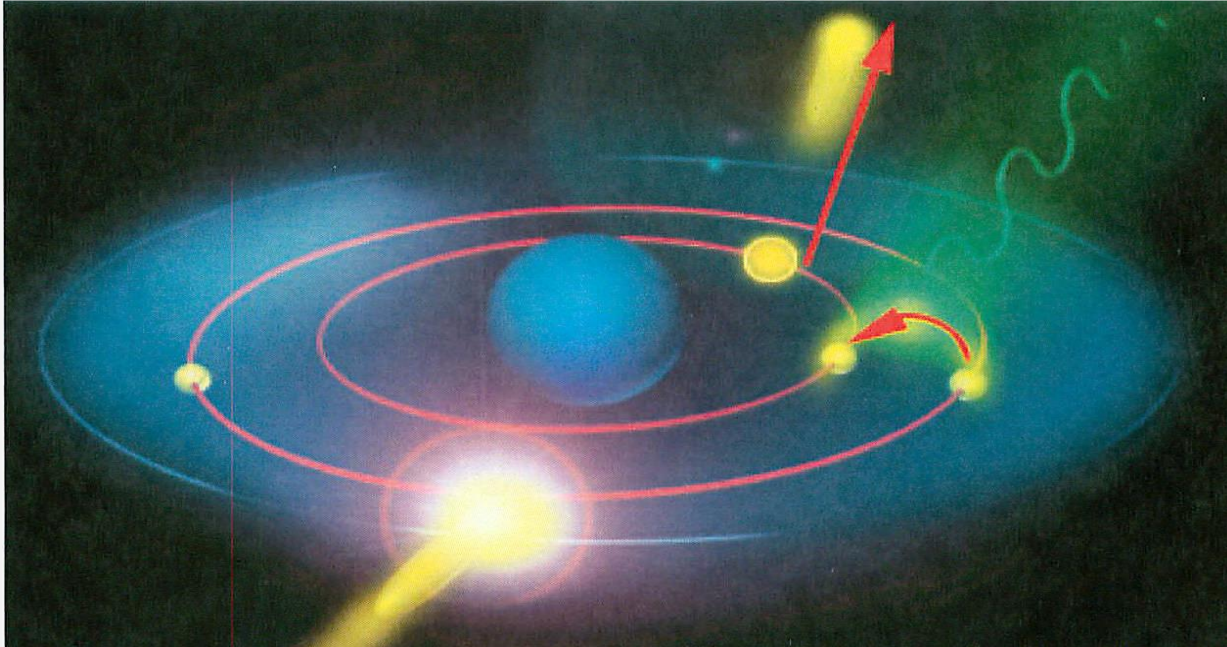
ATOM structure



$$\left\{ \begin{array}{l} K = \text{max } 2 e^- \\ L = \text{max } 8 e^- \\ 1 \times 1 = \text{max } 18 e^- \\ \vdots \end{array} \right.$$

Brehmstrahlung



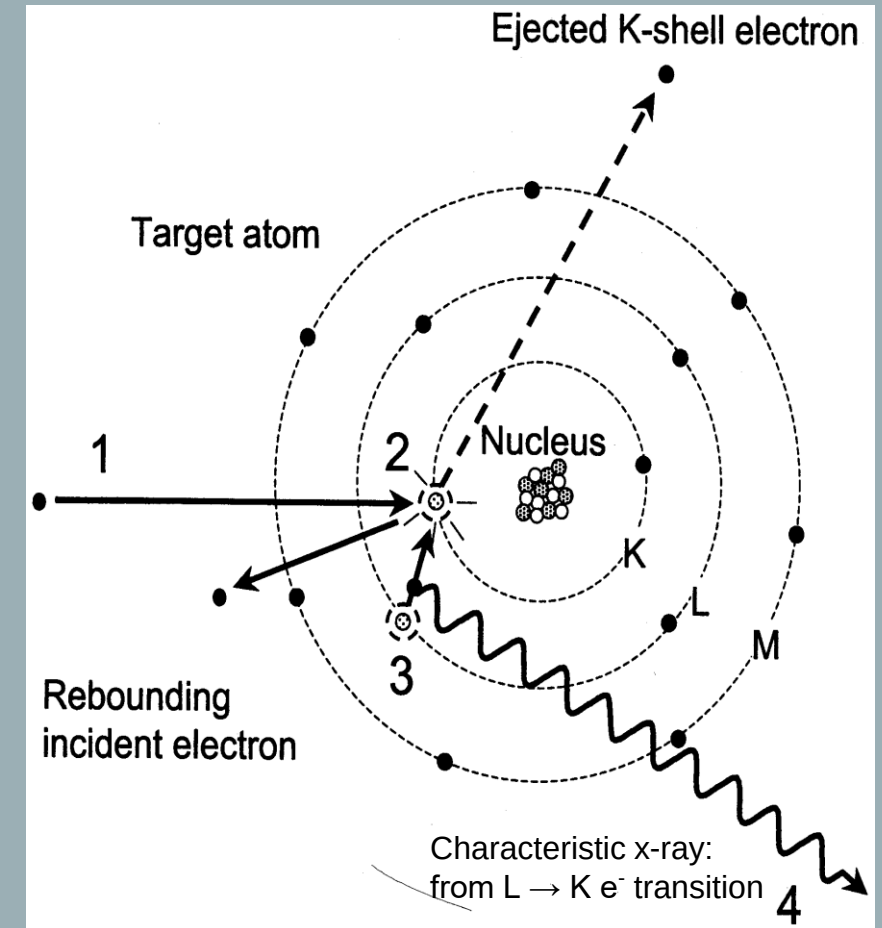
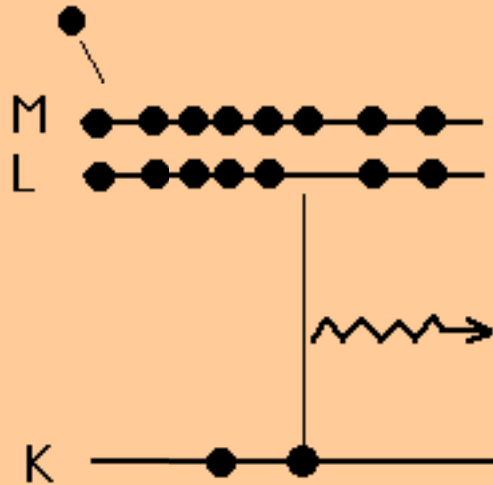


Characteristic

X-rays

Mechanism

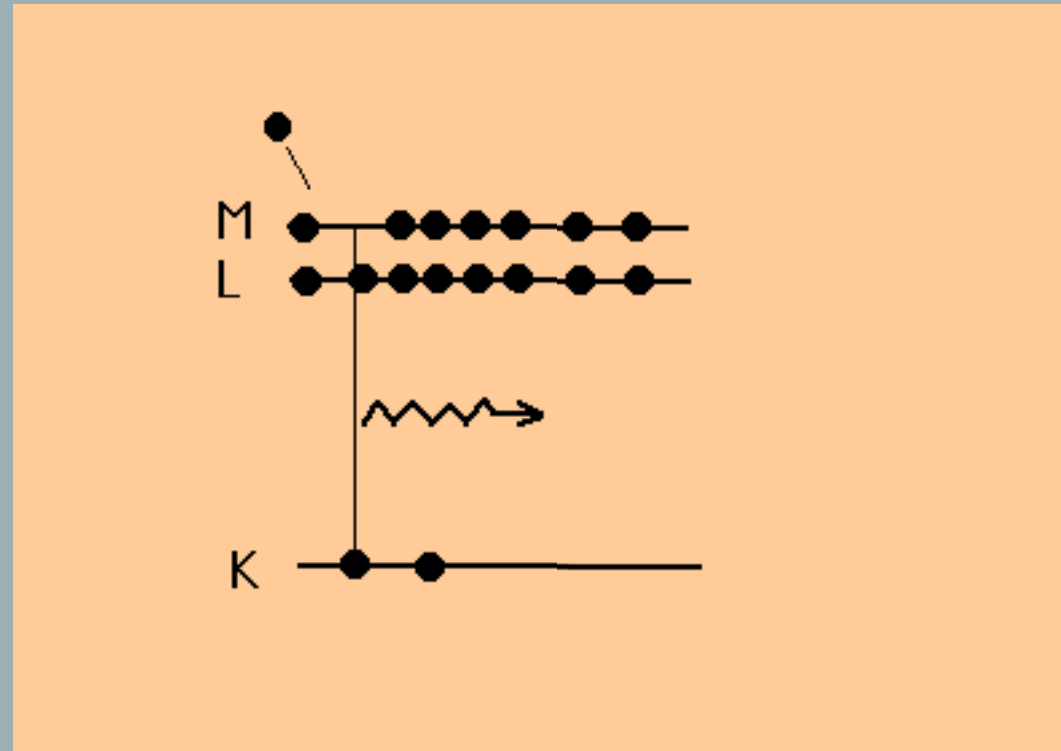
Decelerating charges give off radiation



X-rays

Mechanism

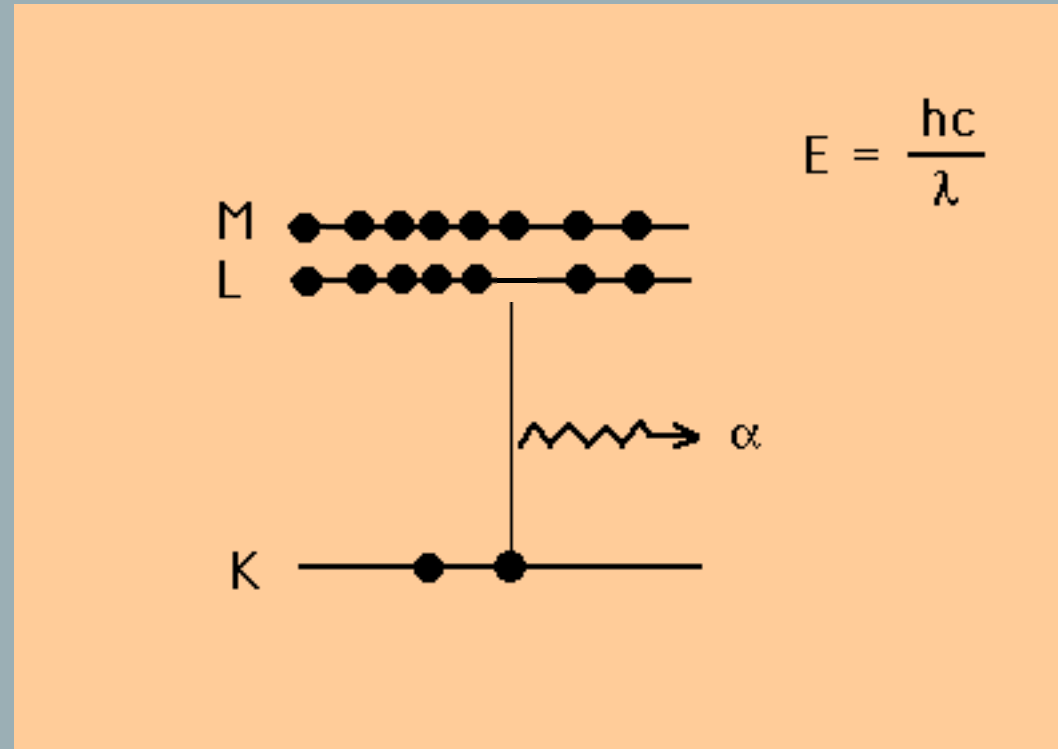
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X-rays

Mechanism

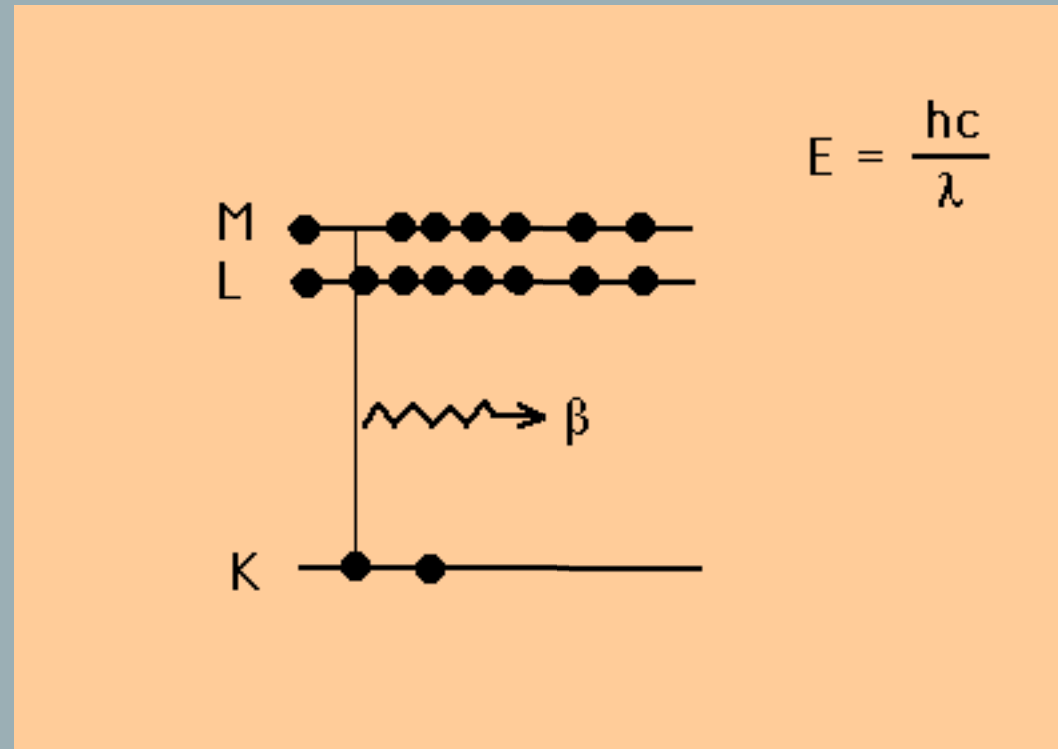
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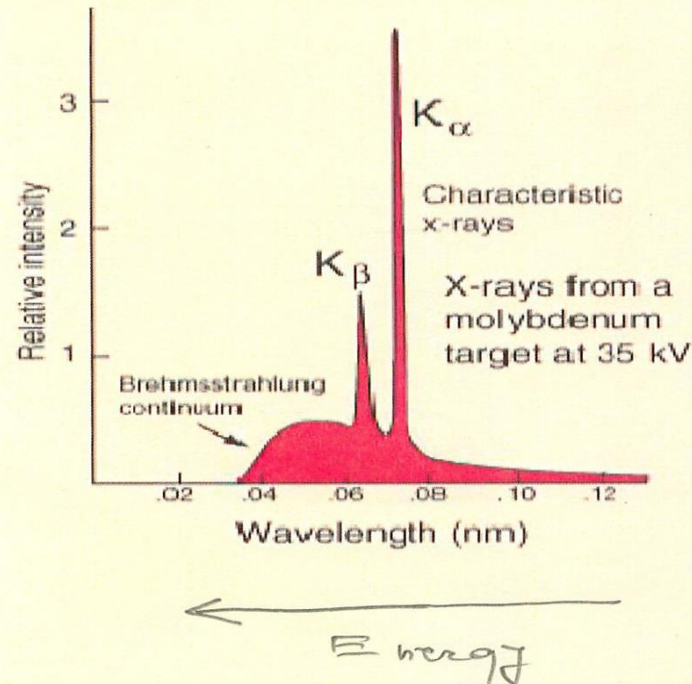
X-rays

Mechanism

Decelerating charges give off radiation



Characteristic X-Rays

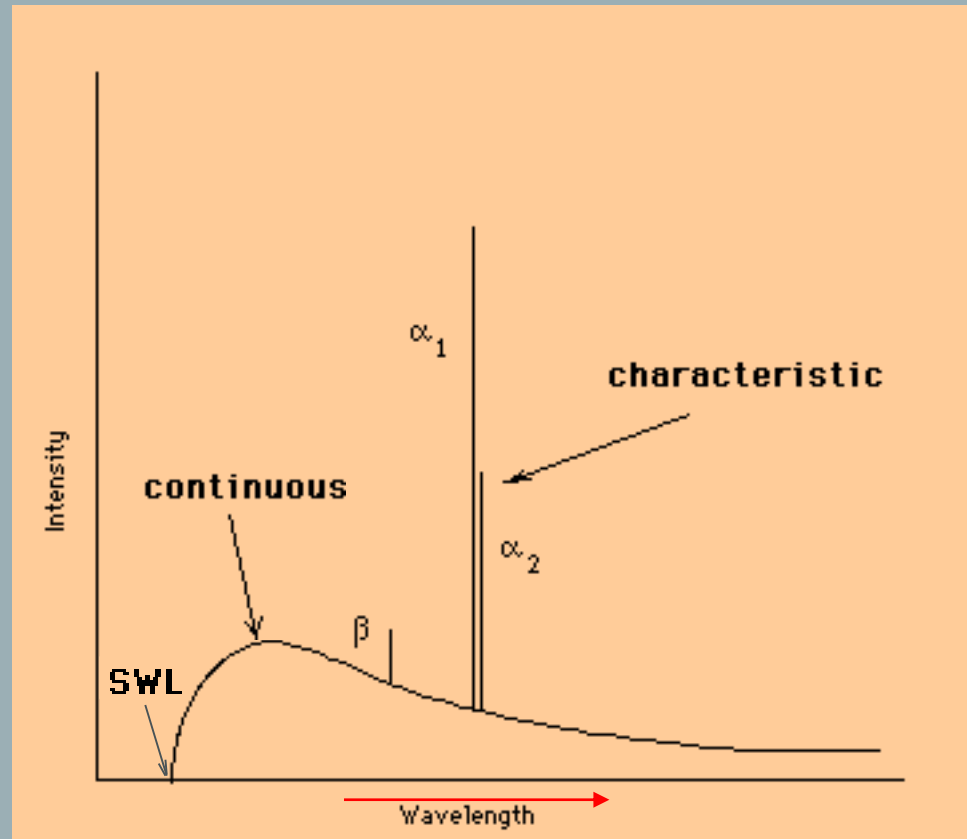


Characteristic x-rays are emitted from heavy elements when their electrons make transitions between the lower atomic energy levels. The characteristic x-rays emission which shown as two sharp peaks in the illustration at left occur when vacancies are produced in the $n=1$ or K-shell of the atom and electrons drop down from above to fill the gap. The x-rays produced by transitions from the $n=2$ to $n=1$ levels are called K-alpha x-rays, and those for the $n=3 \rightarrow 1$ transition are called K-beta x-rays.

Transitions to the $n=2$ or L-shell are designated as L x-rays ($n=3 \rightarrow 2$ is L-alpha, $n=4 \rightarrow 2$ is L-beta, etc.). The continuous distribution of x-rays which forms the base for the two sharp peaks at left is called "bremsstrahlung" radiation.

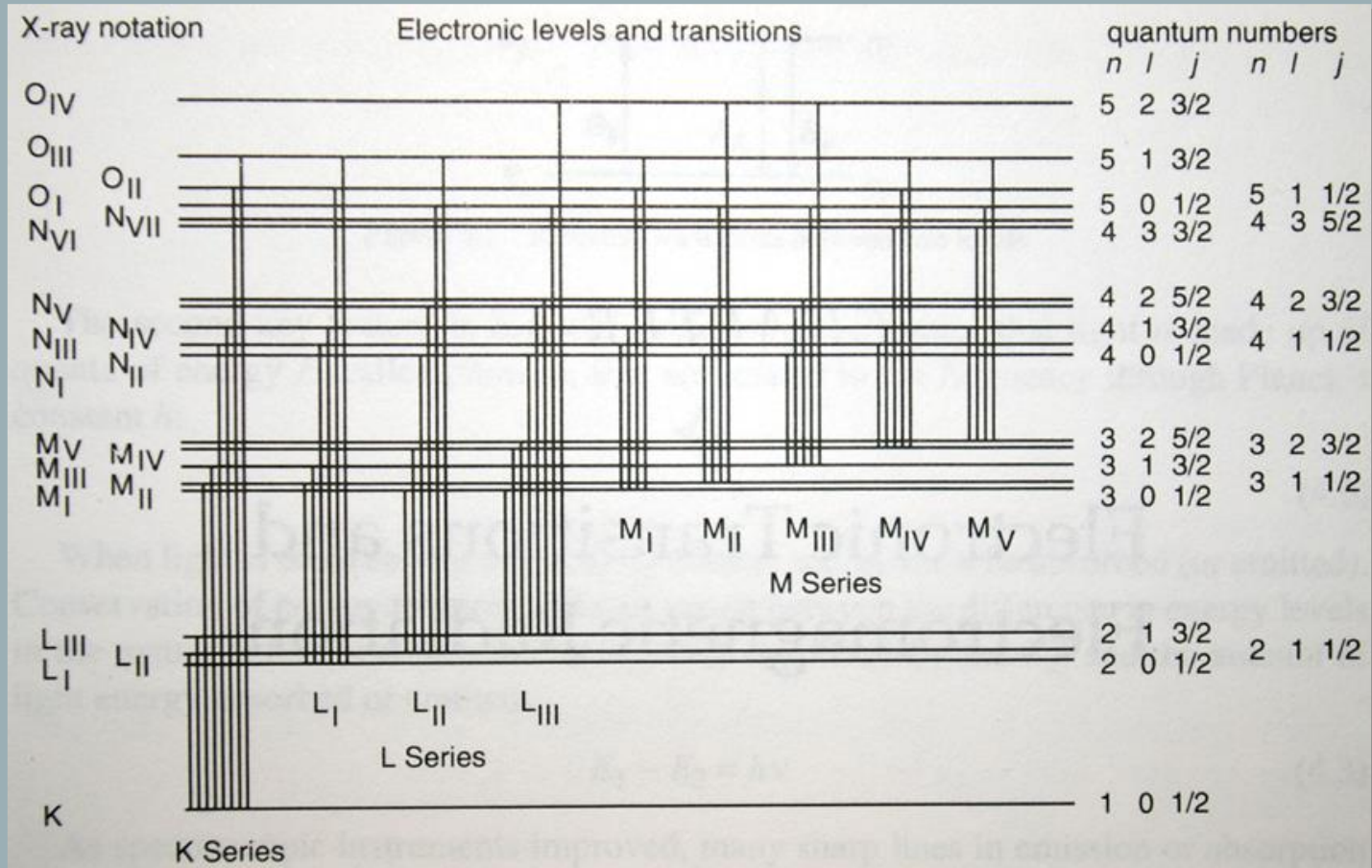
X-rays

Typical tube spectrum



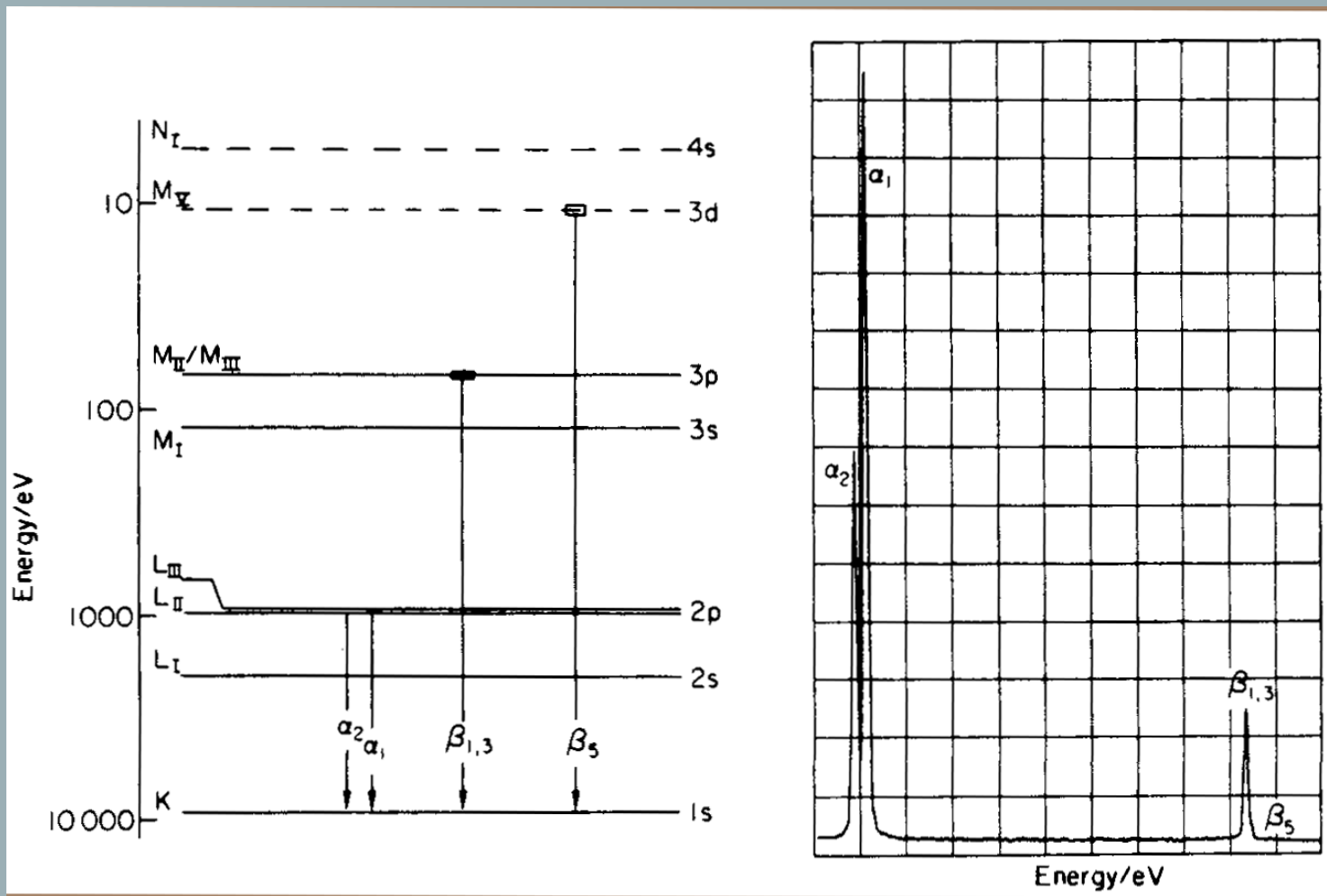
X-rays

More electron transitions



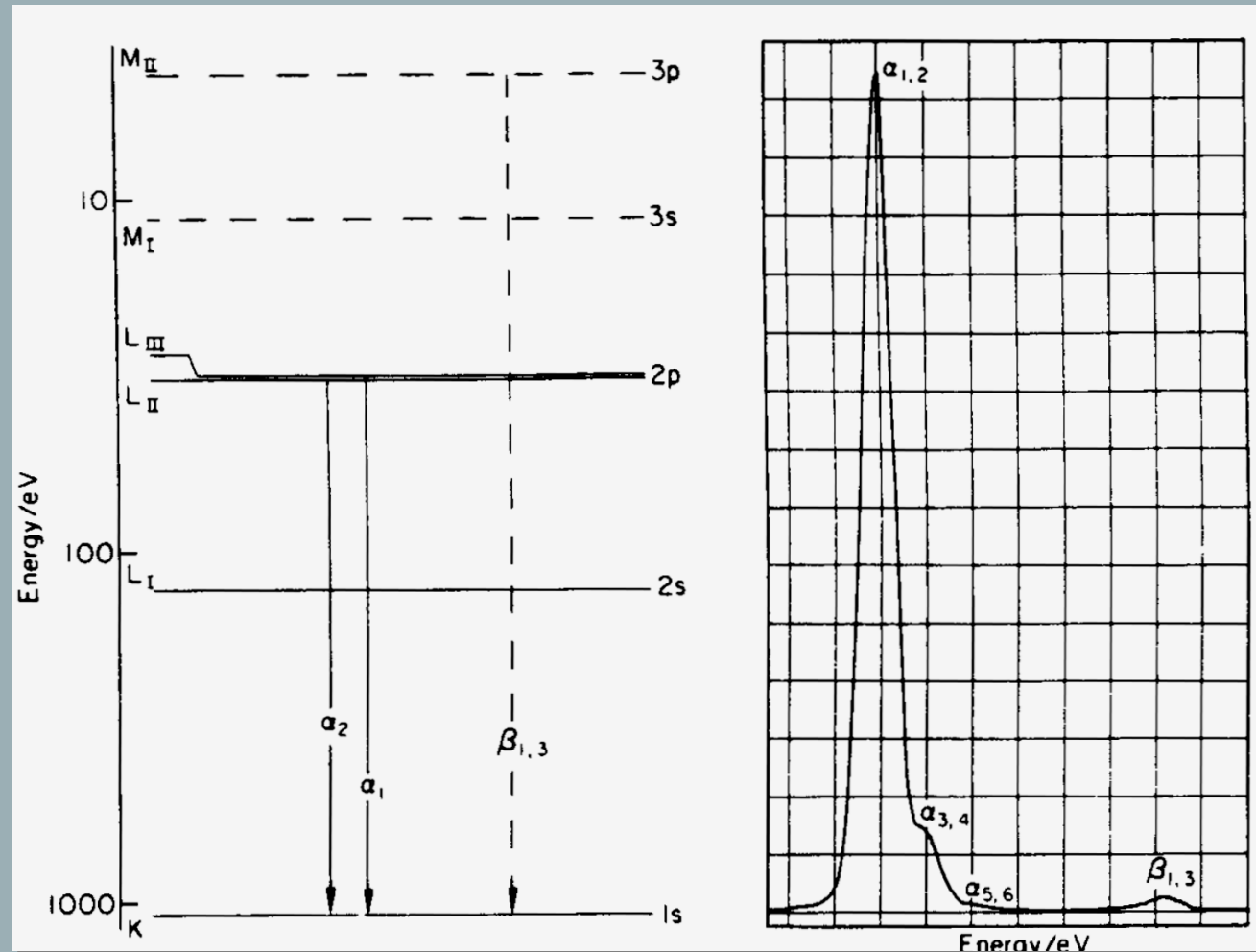
X-rays

Cu spectrum



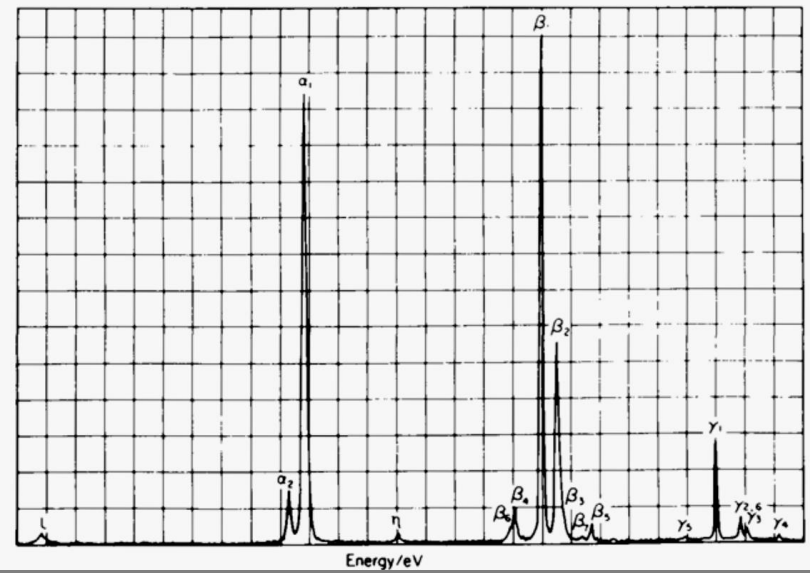
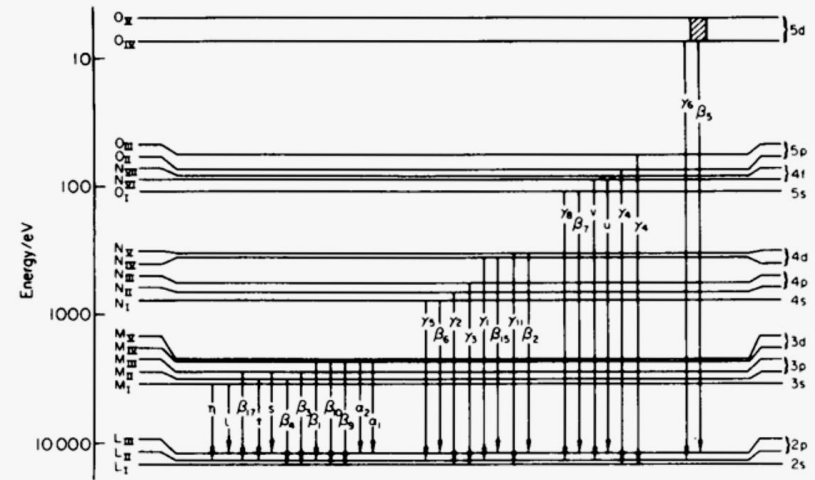
X-rays

Al spectrum



X-rays

Au L spectrum



X-ray sources

Sealed tubes - Coolidge type
common - Cu, Mo, Fe, Cr, W, Ag



Some commonly used K wavelengths

Element	$K\alpha$ (weighted average)*	$K\alpha_2$ strong	$K\alpha_1$ very strong	$K\beta_1$ weak
Cr	2.29100	2.293606	2.28970	2.08487
Fe	1.937355	1.939980	1.936042	1.75661
Co	1.790260	1.792850	1.788965	1.62079
Ni	1.659122	1.661747	1.657910	1.500135
Cu	1.541838	1.544390	1.540562	1.392218
Mo	0.710730	0.713590	0.709300	0.632288

* $K\alpha_1$ is given twice the weight of $K\alpha_2$.

Characteristic L Lines of Tungsten

Line	Relative intensity	Wavelength
$L\alpha_1$	Very strong	1.47639
$L\alpha_2$	Weak	1.48743
$L\beta_1$	Strong	1.281809
$L\beta_2$	Medium	1.24460
$L\beta_3$	Weak	1.26269
$L\gamma_1$	Weak	1.09855

$$\lambda_{K\alpha} = (2\lambda_{K\alpha_1} + \lambda_{K\alpha_2})/3$$

X-ray sources

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X-ray sources

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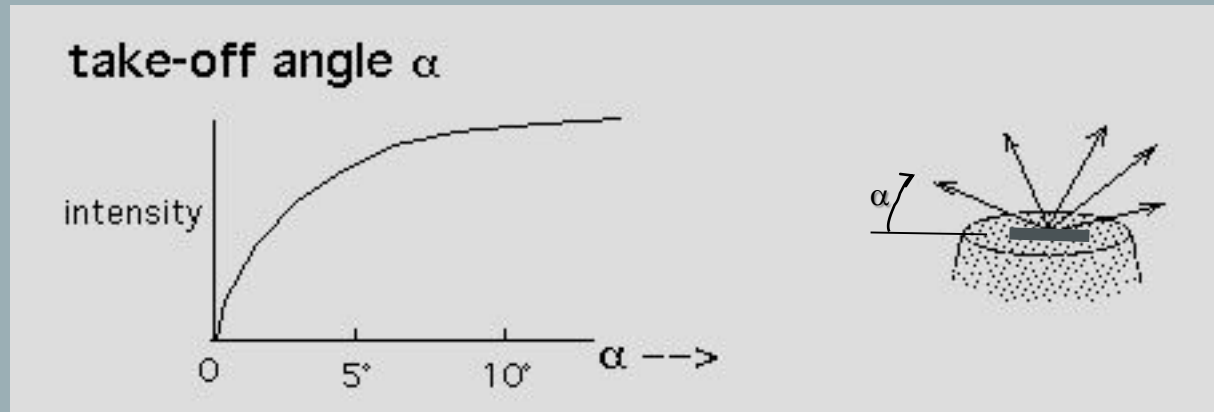
intensity limited by cooling requirements (2-2.5kW)

(~99% of energy input converted to heat)



X-ray sources

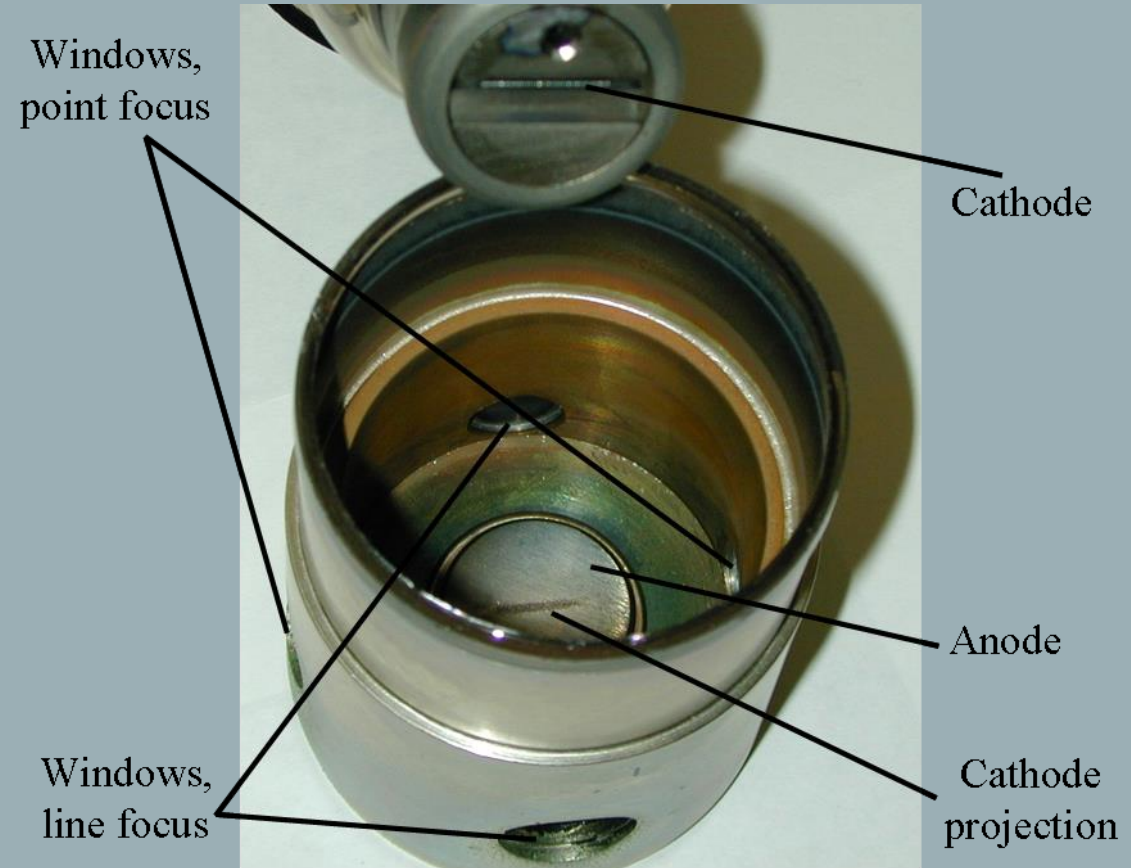
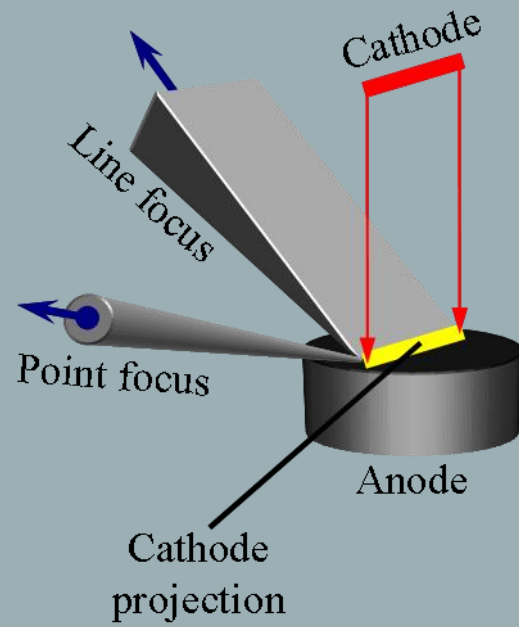
Intensity changes with take-off angle



But resolution decreases with take-off angle

focal spot - normal focus 1x10 mm --> 0.1x1mm (line) $\alpha = 6^\circ$
1x1mm (spot)
fine focus 0.4 x 0.8 mm --> 0.04 x 0.8 mm (line)
0.4 x 0.8 mm (spot)

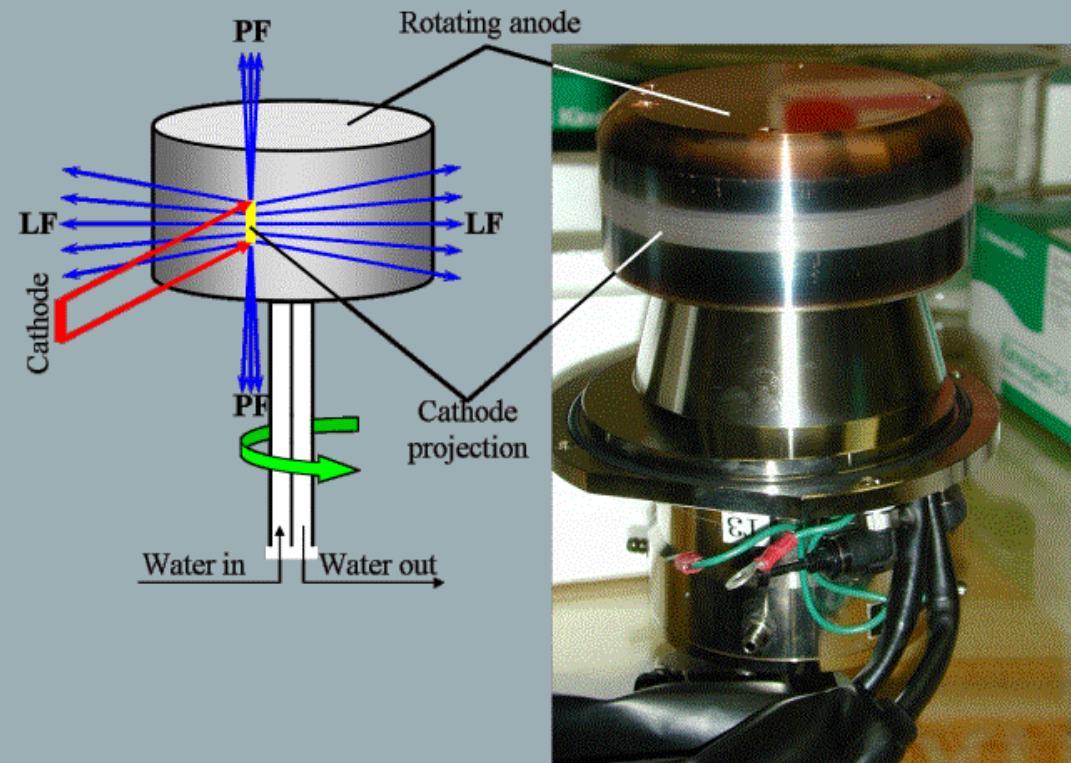
X-ray sources



Other X-ray sources

Rotating anode

high power - 40 kW
demountable
various anode types



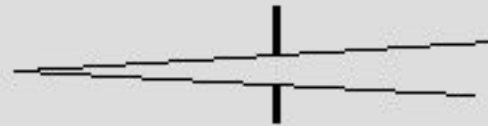
Beam conditioning

Collimation

parallel beam



divergent beam



Beam conditioning

Monochromatization

β -filters – materials have atomic nos. 1 or 2 less than anode

50-60% beam attenuation

placing after specimen/before detector

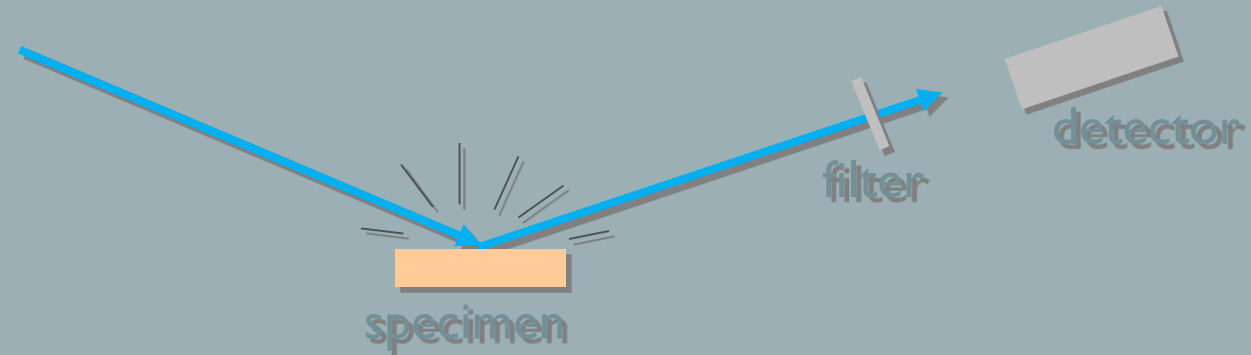
specimen fluorescence

allows passage of high intensity & long
white radiation

filters most of

wavelength

X-rays



Beam conditioning

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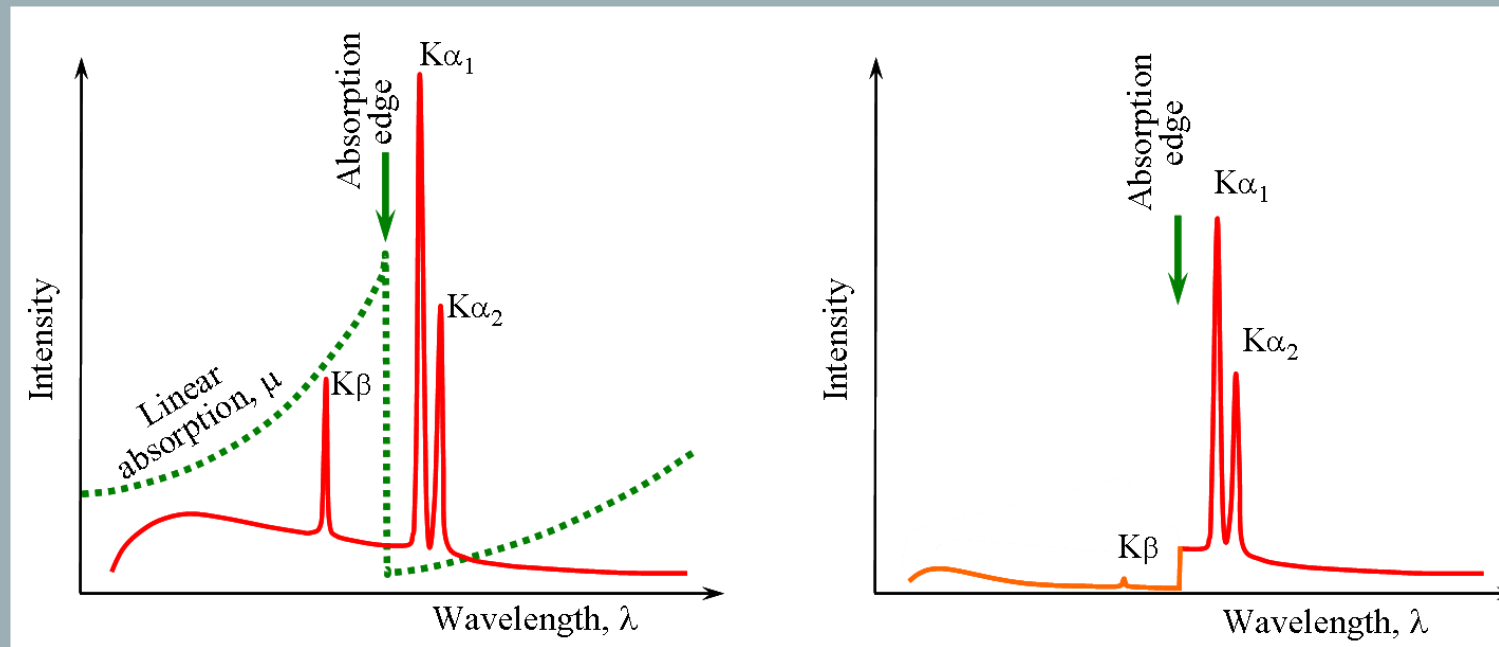
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Beam conditioning

Monochromatization

Crystal monochromators – LiF, SiO₂, pyrolytic graphite
critical – reflectivity ex: for MoK_α, LiF 9.4%
graphite 54 %

Beam conditioning

Monochromatization

Crystal monochromators – LiF, SiO₂, pyrolytic graphite

critical – reflectivity ex: for MoK_α, LiF 9.4%
graphite 54 %

resolution – determines peak/bkgrd ratio &
spectral purity best - Si = 10"
graphite = 0.52°



Applications

Higher resolution

Chest Xrays rib etc

Bones

Dental X rays

Fluoroscopy, mammography, angiography

CT